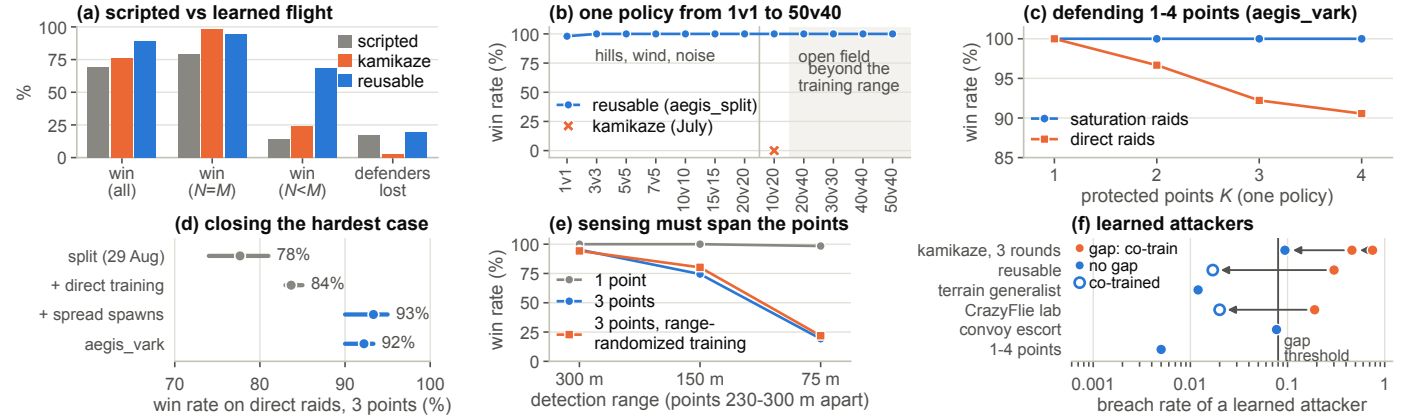

(a) scripted vs learned flight
%
100
75
50
25
0
scripted
kamikaze
reusable
win (all)
win (N=M)
win (N<M)
defenders lost
(b) one policy from 1v1 to 50v40
win rate (%)
hills, wind, noise
open field
beyond the training range
reusable (aegis_split)
kamikaze (July)
1v1
3v3
5v5
7v5
10v10
15v15
20v20
10v20
20v40
30v30
40v40
50v40
(c) defending 1-4 points (aegis_vark)
win rate (%)
saturation raids
direct raids
protected points K (one policy)
(d) closing the hardest case
split (29 Aug)
78%
+ direct training
84%
+ spread spawns
93%
aegis_vark
92%
win rate on direct raids, 3 points (%)
(e) sensing must span the points
win rate (%)
1 point
3 points
3 points, range-randomized training
300 m
150 m
75 m
detection range (points 230-300 m apart)
(f) learned attackers
kamikaze, 3 rounds
reusable terrain generalist
CrazyFlie lab convoy escort
1-4 points
gap: co-train
no gap
co-trained
gap threshold
breach rate of a learned attacker